

Passive Pre-Association Wi-Fi Access-Point Selection

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Wi-Fi AP Selection Research Project

Abstract— A Wi-Fi client in overlapping coverage must select an access point (AP) before it can directly measure the service that each candidate would provide. Signal strength describes link quality, but not contention or load, and may therefore favor a nearby but congested AP. This project investigates whether a client can make a better decision using only passive observations collected before association, without probe requests or assistance from the network. This report defines the experimental foundation of that investigation: a controlled ns-3 simulator that creates multi-AP deployments, records decoded frames and physical carrier-sense activity, and replays each fixed topology once per candidate AP, changing only the AP the candidate joins. Every replay of a topology therefore shares one identical pre-association observation while yielding its own measured throughput. The simulator separates physical-topology randomness from radio randomness and produces a structured choice-set dataset. The resulting method supports a reproducible comparison between conventional signal-based selection and selection based on richer passive evidence.

I. PROBLEM FORMULATION

Throughput depends on channel contention, competing stations, transmission rates, retries, and the airtime consumed by neighboring basic service sets. Received signal strength reflects none of them directly. Those effects do, however, leave traces a client can hear without transmitting: the frames themselves, and the time the medium spends busy. This work asks whether such traces are enough: *can a client choose the AP that will provide the greatest throughput using only what its radio can passively observe before association?*

Existing load-aware methods [1], [2] make their load estimate after association, from AP-provided reports, or from a network controller. A client making its first choice has none of these: only beacons and some data frames, in which AP distance and offered traffic appear only through their effects.

Let $\mathcal{A} = \{a_1, \dots, a_N\}$ be the APs in one deployment and let O be the candidate station's passive observation before it joins any AP. If y_i is the throughput obtained after joining a_i , the desired decision is

$$a^* = \arg \max_{a_i \in \mathcal{A}} y_i, \quad (1)$$

using a decision function whose inputs are derived only from O . In a live network only one y_i is ever measurable: the station joins a single AP and observes that AP's throughput, while the values of y_j for every AP it did not join are counterfactual. The simulator recovers all N values by replaying the same scenario

once per target AP, changing only the AP the candidate joins. Every choice set therefore carries its true a^* .

The replay is sound only if O is identical across all N runs of a scenario. If any part of the observation varied with the AP eventually joined, a decision function could reach the right answer through a simulation artifact rather than through the network. This is checked rather than assumed. The assumptions that do remain are stated next.

II. EXPERIMENTAL SCOPE AND ASSUMPTIONS

Simulator 1 isolates the relationship among path quality, background load, and passive evidence. It does not attempt to reproduce every feature of a deployed WLAN. Table I states the principal assumptions so that conclusions can be interpreted within the intended scope.

TABLE I
CONTROLLED ASSUMPTIONS OF SIMULATOR 1.

Aspect	Current assumption
Deployment	Static, single-floor, no walls; one stationary candidate and stationary background stations.
Radios	Homogeneous IEEE 802.11n devices at 5 GHz with 20 MHz channels and Minstrel-HT rate control [5].
Propagation	Yans Wi-Fi channel; log-distance loss (exponent 3.0, 46.6777 dB reference loss at 1 m) followed by Nakagami fading.
Traffic	Uplink UDP; constant background offered rates and a saturated candidate. Wired backhaul is not a bottleneck.
Observation	Passive and client-side; no probe request, association, or network-provided load report during feature collection.
Target	Mean candidate uplink throughput from successful association until the simulation stops; failed association is assigned zero.

The environment is static and open, so it excludes mobility, walls, hardware diversity, and application demand that varies over time. The label is uplink UDP throughput, so it says little about TCP goodput, downlink performance, or latency.

The choice set contains only APs discovered during the projected scan. In passive mode, discovery requires at least one decoded beacon, and observations with fewer than two discovered APs are excluded because no selection decision exists.

III. SIMULATOR DESIGN

The simulator is built on ns-3.45 and its Wi-Fi models [6]. APs sit 30 m apart in a fixed layout: two form a pair, three an equilateral triangle, and four, six, and eight fill 2×2 , 3×2 , and 4×2 grids. Channels 36, 40, 44, and 48 are assigned in

AP-index order and reused once a deployment exceeds four APs. Because the layout is fixed as well, co-channel separation takes a single value per deployment size — 30 m at eight APs, 42 m at six, and no reuse at or below four — which ties channel reuse to AP count. Section VIII returns to this.

Each scenario draws 2, 3, 4, 6, or 8 APs and 2, 3, or 4 background stations per AP. A quarter of scenarios spread those stations uniformly. The rest place one or more hotspot APs, with fewer hotspots more likely and the count capped at $\lceil N_{AP}/3 \rceil$. Around those hotspots 70% of background stations fall in 10 m discs, and the remainder cover the footprint. Mixing uniform and deliberately clustered regimes follows common practice in WLAN studies [3], [4], and matters here because under uniform load every AP is about equally busy and the choice collapses back to signal strength. Each station associates with its nearest AP, and every background station in a scenario offers the same load, drawn uniformly at random from 1.0, 2.5, 4.0, and 6.0 Mbit/s. Candidate positions are drawn to keep the decision non-trivial: 70% near a boundary between neighboring APs, 30% at 4–16 m from a single AP. These are sampling policies rather than radio parameters, and they are recorded with the dataset because they set its difficulty.

Background traffic starts at $t = 1$ s. Passive recording runs until $t = 5.5$ s, covering 4.5 s of loaded traffic; the candidate associates at $t = 6$ s and the run ends at $t = 18$ s. The 0.5 s gap between recording and association is a guard: ns-3 performs target-dependent polling just before a join, and without the gap that polling would land inside the observation window. After joining, the candidate offers 100 Mbit/s in 1250-byte packets, which keeps it backlogged so that measured throughput reflects the service the network gives it rather than the limits of its own traffic generator.

A. Matched Counterfactual Replays

For each sampled topology and radio seed, the experiment runs once per target AP. AP and station positions, background traffic, the candidate’s position, and every pre-association device stay fixed; only the AP selected at $t = 6$ s changes. A `topologySeed` controls background-station placement and an independent `rngSeed` controls fading, contention, and rate adaptation, with five radio seeds generated per topology. Separating the two streams is what allows stochastic radio behavior to be repeated without the physical deployment shifting underneath it.

B. Validation

Validation runs before any model is fitted, so that a good accuracy number cannot mask a defective data-generating process.

Three properties carry the experimental design, and each is checked automatically. Repeating a seed reproduces throughput exactly, so any run can be regenerated. Changing only the radio seed varies the outcome while every physical position stays fixed. And the recorded frame and channel-busy traces are byte-identical across all target-AP replays of a matched scenario —

the check that enforces the constant O required in Section I, and the most important of the three.

These establish internal consistency, not that the simulator behaves like a real network.

IV. OBSERVATION AND DATASET CONSTRUCTION

A. Recording and Projection

The candidate carries one receive-only scanner PHY per occupied channel, plus a separate PHY reserved for association. The scanner PHYs never transmit and never associate; the association PHY is parked on unused channel 149 until the decision time, so the AP the candidate will eventually join cannot alter what is heard, or consume random draws, while the observation is being collected. At $t = 5.5$ s the scanner PHYs move to the parking channel as well.

Watching every channel at once is not a claim that real clients have several radios. It is a recording device: it captures a superset of what any single radio could have heard, from which a realistic single-radio observation is cut afterwards.

Two traces are recorded. The monitor-sniffer trace holds every frame a scanner decodes, with time, frequency, BSSID, transmitter, frame category, beacon and retry flags, length, RSSI, noise, airtime, and PHY rate. The PHY-state trace gives the fraction of each 102.4 ms beacon interval in which carrier sense marked the channel busy. This clear-channel-assessment (CCA) measurement registers energy a frame-only view misses: interference, and transmissions too distant to decode. Real chipsets expose such a counter, though whether a driver surfaces it varies by platform.

The superset is then projected onto one passive radio. That radio rotates through the four channels in random order, dwelling 110 ms on each with 0.15 ms to retune, and repeats the cycle for as much of the window as it fits, so every channel is revisited many times before the decision. Frames and CCA intervals falling outside a dwell are discarded. A 110 ms dwell spans just over one 102.4 ms beacon interval, so each visit yields about one beacon from every AP on that channel. RSSI is then perturbed by uniform error within ± 3 dB and quantized to 1 dB. Both steps run after ns-3 from a recorded scan seed, so scan policy and measurement noise can be varied without repeating a simulation.

B. From Trace to Training Data

The projected trace is a long list, one entry per decoded frame, with the channel-busy series recorded alongside it. Turning that list into training data means collapsing it once per discovered AP. For a given AP the builder first selects the parts of the trace that bear on it — frames sent by or to its BSSID, and activity on the channel it occupies — which yields a shorter list, not yet a single description. It then reduces that shorter list to a fixed-size summary of the AP, and it is the summary, not the frames, that a model ever sees. Four discovered APs therefore give four summaries over identical evidence, differing only in which AP each describes and in the throughput that AP delivered when it was the target of a replay.

That reduction is carried out twice, in two different ways. The first produces a fixed vector of summary statistics — these are the `feat_` columns, and they are hand-chosen. The second keeps time: the window is cut into equal bins and each measurement is recorded separately in every bin, producing a sequence per option instead of a vector. Tabular models are trained on the first, the transformer on the second, and Section V compares them.

The summary vector carries three kinds of feature. AP-level features describe that BSS itself: RSSI, beacons, observed clients, frames, airtime, retries, and PHY rates. Channel-level features describe the option’s channel as a whole, decoded traffic and CCA busy time together. Relative features place the option against its rivals: how many options there are, its RSSI rank, and its margin over the best alternative.

Ground truth is kept but quarantined. True distance, configured load, station count, positions, and seeds are all recorded, because they are needed to check that the observable features behave and to break results down by scenario type. None may reach a model. The two are separated by name: only `feat_` columns are admissible inputs, everything with a `gt_` prefix is not, and the loader enforces the split rather than trusting discipline.

Rows are grouped twice over. Each physical topology carries a `topology_id`, and each of its five radio-seed realizations carries its own `group_id`. Rows sharing a `group_id` were produced from one identical observation and differ only in which AP was joined. Partitions must therefore be cut by `topology_id`: the five realizations of a layout are near-duplicates of each other, and letting them fall on both sides of a train/test split would inflate every result.

V. MODELS AND EVALUATION PROTOCOL

Two observations are compared. The *rotating* observation is the single-radio projection of Section IV, in which one radio cycles through the channels for the whole window. The *continuous* observation keeps the parallel-scanner superset instead, so every channel is watched at once. One radio cannot do this, so the continuous condition is an upper bound on what observation could provide rather than a deployable alternative. Both carry the same RSSI realism, and both are binned into per-option sequences.

The dataset holds [–] AP options across [–] observations of [–] topologies, a median of [–] discovered APs per decision. Choosing the best AP rather than at random is worth [–] against [–] Mbit/s on average, and the mean spread between the best and worst option in a choice set is [–] Mbit/s.

Four families of decision function are compared. Heuristics score an option directly: strongest RSSI, least busy channel, and RSSI minus channel-busy fraction. Tabular models take one feature vector per option, under three encodings of the same sequence — its window means, a set of summary statistics per measurement (mean, standard deviation, minimum, maximum, last value, and least-squares slope), and the whole sequence flattened with bin order preserved. A ridge regression takes the flattened sequence as a linear control. The transformer encodes

each AP’s sequence with a learned positional embedding, then compares APs through a permutation-invariant set stage, and is trained twice: once to predict throughput, once to rank options directly.

Every model sees the same five splits, cut by `topology_id` and stratified by AP count. Metrics are top-1 accuracy with a 0.5 Mbit/s tie tolerance, mean regret in Mbit/s against the oracle choice, and R^2 where a model predicts throughput. Significance is a paired *t*-test across the five splits.

VI. RESULTS

Table II gives top-1 accuracy for every model under both observations. Heuristic reference points are identical in the two columns, since they read the same features: random selection reaches [–], least busy channel [–], strongest RSSI [–], and RSSI minus busy [–].

TABLE II
TOP-1 ACCURACY BY MODEL AND OBSERVATION. WITHIN A COLUMN EVERY MODEL RECEIVES IDENTICAL INPUT.

Model	Rotating	Continuous	Gain
Ridge, flattened sequence	[–]	[–]	[–]
MLP, flattened sequence	[–]	[–]	[–]
MLP, summary statistics	[–]	[–]	[–]
GBM, summary statistics	[–]	[–]	[–]
GBM, flattened sequence	[–]	[–]	[–]
Transformer	[–]	[–]	[–]

Three comparisons are read off this table. The first is which model wins under each observation, and by how much, given that every model in a column receives identical input. The second is what widening the observation is worth, model by model, which the gain column reports. The third is whether the winner changes between the two columns.

Two ablations accompany it. Giving every tabular model the mean of the other options and its own deviation from that mean supplies the cross-option comparison the transformer’s set stage performs, and isolates how much of any transformer margin is attributable to it. Shuffling the time bins within each option, leaving bin position intact, removes temporal order while preserving everything else, and isolates aggregation from tracking.

VII. DOES THE SEQUENCE CARRY ANYTHING?

The matched design allows the label’s variance to be split directly. Across cells of one topology paired with one target AP, each realized under five independent radio seeds, the intraclass correlation separates what is fixed when the topology is drawn from what varies with the radio alone. A high value bounds what any time-sensitive model can contribute, because it says the outcome is settled before the observation begins; a low one leaves room for the trajectory to matter. The measured value is [–], against a between-cell variance of [–] and a within-cell variance of [–].

The shuffle ablation is the independent check on the same question. If order carries information, destroying it must cost accuracy; if the sequence is repeated sampling of a stationary

environment, it will not. The two instruments are reported together because agreement between them is what makes either conclusive.

VIII. WHAT SIMULATOR 2 MUST CHANGE

The requirement is measurable rather than a matter of judgement. Recomputing the intraclass correlation on a successor dataset is a direct test of whether the question can be asked at all: unless it falls well below 0.9, no amount of model engineering will make a temporal model useful, because there will again be nothing for it to track. What must change is therefore not the model but the environment — load that arrives and departs, interference that appears part-way through the window, stations that join and leave — and it must be visible in the passive trace while remaining invisible to a window average.

Four smaller decisions follow from the present work. Channel assignment by AP index, over a fixed layout, leaves co-channel separation with no variance and ties reuse to deployment size; sampling the assignment instead would decouple the two. AP spacing of 30 m pairs an obstructed-indoor path-loss exponent with an open-field geometry, and the two should be reconciled. The BSS Load element, which a passive client can read from a single beacon, is absent here and is the strongest baseline this comparison currently lacks. And the restriction to complete silence, rather than to the pre-association period, is a scope choice that could be relaxed to admit probe requests.

One practical consequence deserves separating from the rest. Since observation budget moved accuracy further than any change of model, and one radio cannot watch every channel at once, the lever a real client actually holds is how long it may scan before deciding. Dwell time and number of passes are therefore worth treating as experimental variables in their own right, rather than as fixed properties of the projection.

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